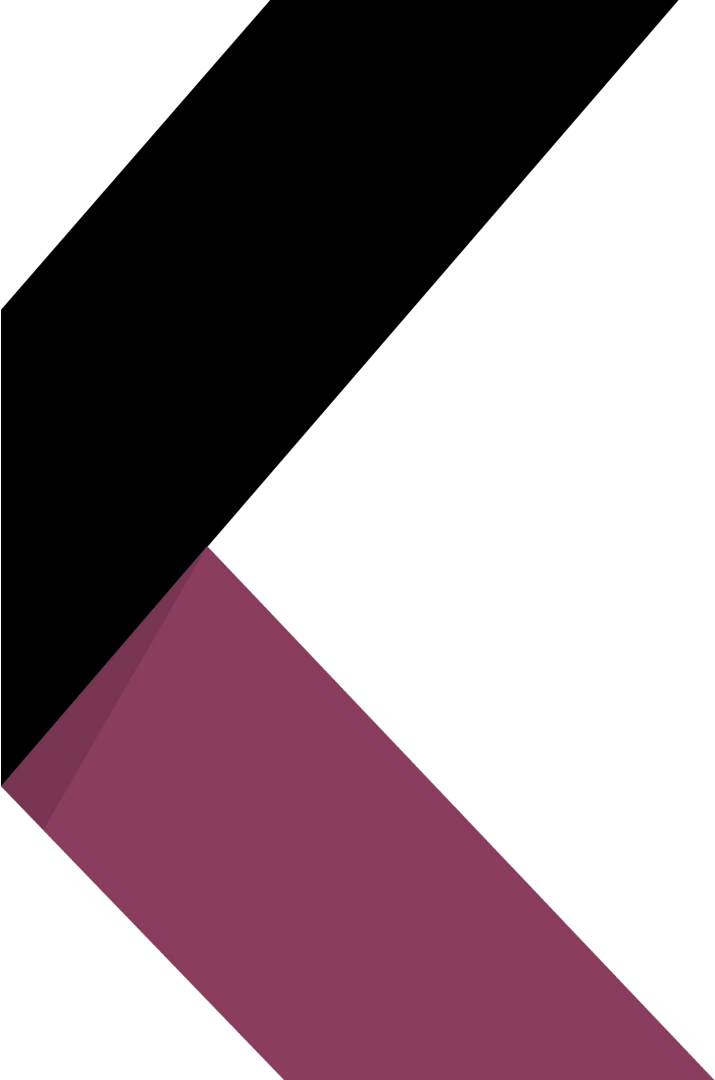


Power converters for DC Machine

The slide features a white background with a large, dark blue rectangle on the right side. A diagonal line divides this rectangle into two triangles: a black triangle on top and a dark blue triangle on the bottom. The title text is positioned to the left of the blue rectangle.



Introduction

Power Converters - Overview

- Although the conventional motor vehicle is well over a century old and was once exclusively powered by heavy mechanical systems, it has evolved with the advent of power electronics (PE). Today, power electronics are incorporated at various levels within all automobiles.
- In EVs, **power electronics is the brain of the powertrain system**. It controls the electric motor and provides the connection to the high-voltage battery. To operate a motor, it is necessary to transfer energy from onboard source(s) to the motor. This energy needs to be transformed in a controlled manner, to create a rotating magnetic field in order to match the load requirements. All these phases (transfer, transformation, and control) are enabled by the power electronic systems.

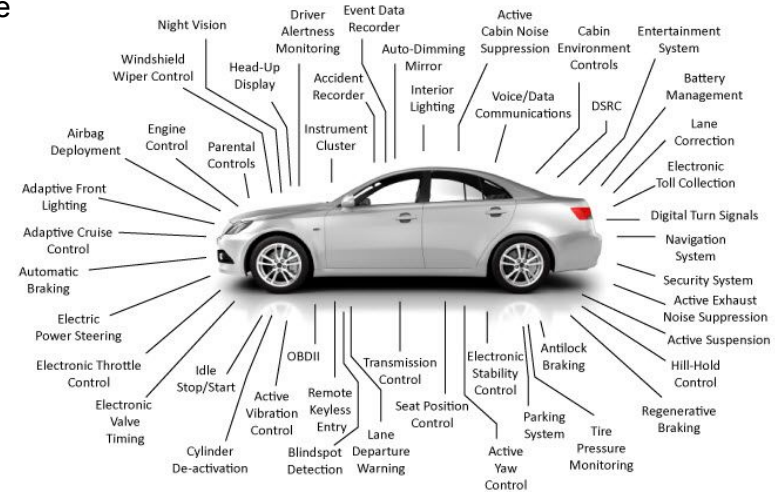


Power Electronics – Definition and applications

Power Electronics deals with the use of electronics for **converting and controlling the flow of electrical energy** from the source to the load in the most efficient way (“power processing”).

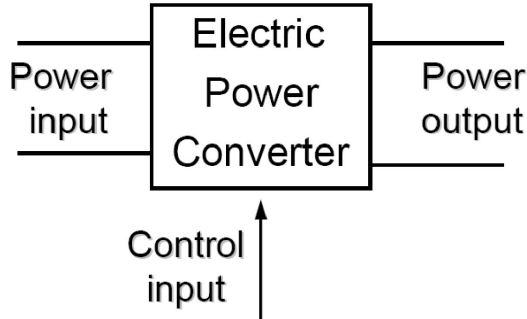
Some automotive applications:

- Electric traction inverters (hybrid and battery-electric drive trains)
- Power management (DC-DC converters, body electronic and auxiliary drives)
- Electrical power steering (EPS)
- Medium-voltage high-power electronics/48V electronics
- Thermal management for power electronics
- Electric vehicle charging

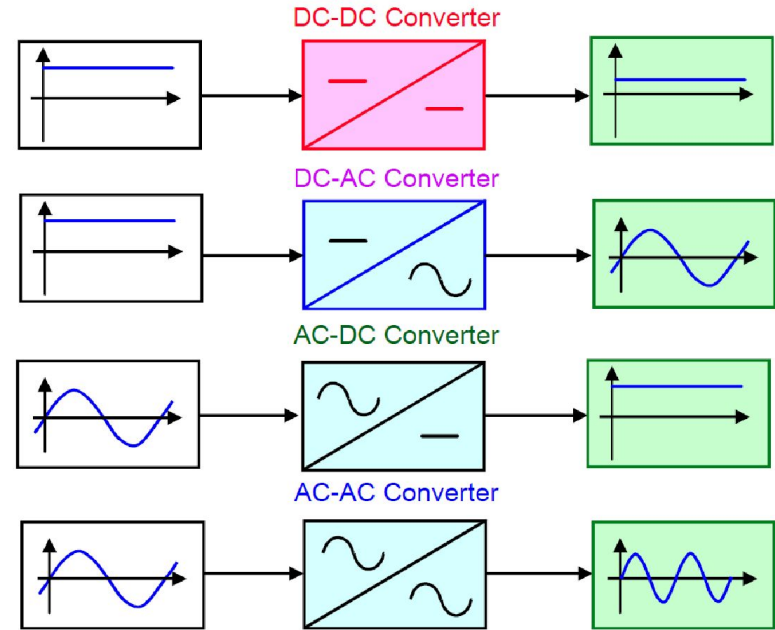




Power converters - Classification



- Dc-dc conversion:* Change and control voltage magnitude
- Ac-dc rectification:* Possibly control dc voltage, ac current
- Dc-ac inversion:* Produce sinusoid of controllable magnitude and frequency
- Ac-ac cycloconversion:* Change and control voltage magnitude and frequency



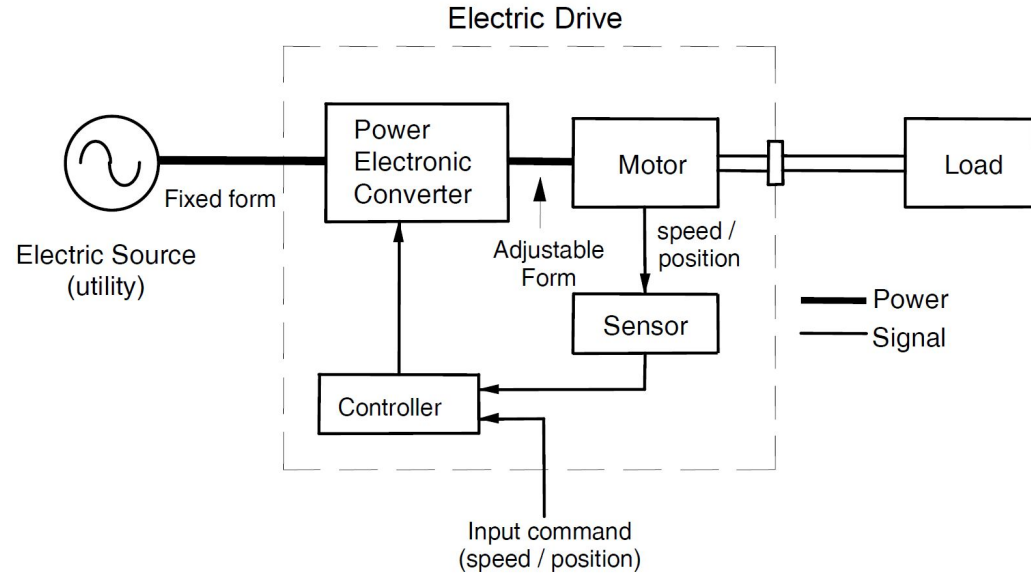


DC Machine - Converters

For DC motors, the choice of converter is primarily dictated by **the nature of the power source**. If this source is alternating current the converters used will be rectifiers that work with natural switching devices, while, if the source is direct current, choppers that are made with forced switching devices are used.

So, the power electronic topologies employed to drive a DC machine are:

- **AC/DC rectifier**
- **DC/DC chopper**





AC/DC rectifiers



AC/DC rectifier - Classification

AC/DC rectifiers can be classified as follows:

1- Single-phase controlled rectifiers

- (i) Single-phase half-wave converter;
- (ii) Single-phase full-wave half-controlled converter;
- iii) Single-phase full-wave fully-controlled converter;
- (iv) Single-phase dual converter.

These are used for **small and medium power motors** up to 75kW (100hp) ratings.

2. Multi-phase controlled rectifiers

- (i) Three-phase half-wave converter;
- (ii) Three-phase full-wave half-controlled converter;
- (iii) Three-phase full-wave fully-controlled converter;
- (iv) Three-phase full-wave dual converter.

Three-phase converters are commonly used in **adjustable speed drives** from about 15 kW up to several thousand kilowatts ratings.

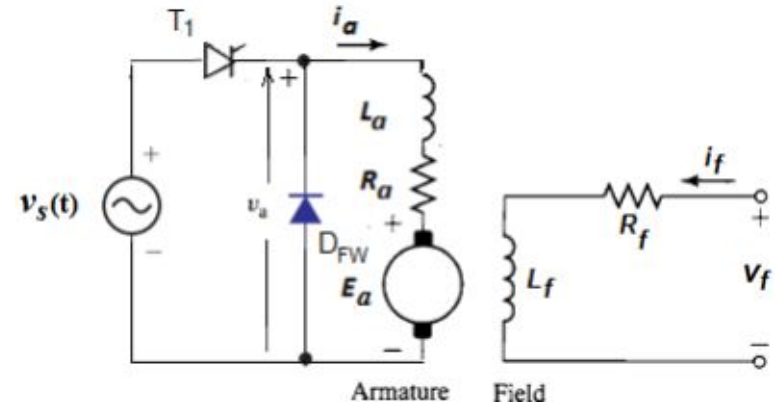


Single-phase half-wave rectifier

This solution is currently used for machines with a power of just over 20 hp, therefore suitable for light traction.

This d.c. drive is very simple, needs only one power switch and one freewheeling diode connected across the motor terminals for the purpose of dissipation of energy stored in the inductance of the motor and to provide an alternative path for the motor current to allow the power switch to commute easily.

However this converter results in **poor motor performance**.





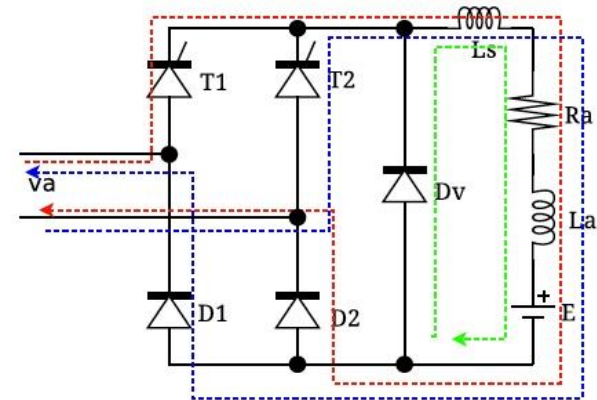
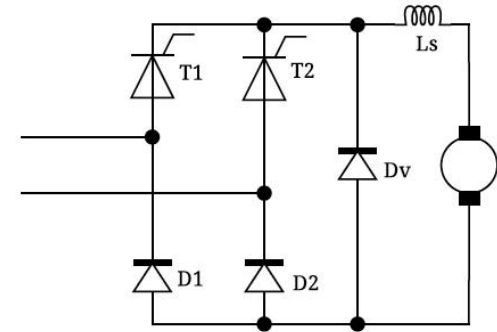
Single-phase full-wave half-controlled rectifier

During the positive half wave T_1 and D_2 are directly biased, and conduction occurs after the command pulse is sent to the thyristor T_1 .

The current flows in the motor through the two components.

The load is then effectively closed in short circuit, the diode D_2 progressively stops conducting and the current passes through the flywheel diode as it represents the path with the lower resistance.

Then for $\omega t = \pi$ the current on T_1 goes to zero and the component switches off. At this point, T_2 and D_1 start conducting. The current conduction proceeds up to the inversion of the half-wave which causes the flywheel diode to polarize again directly. Once the diode conduction is finished, the cycle begins again.

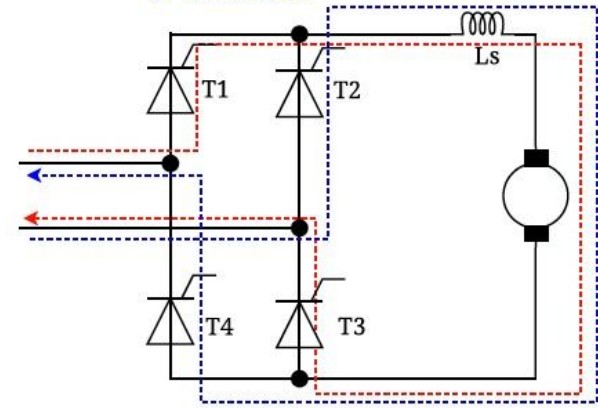




Single-phase full-wave fully-controlled rectifier

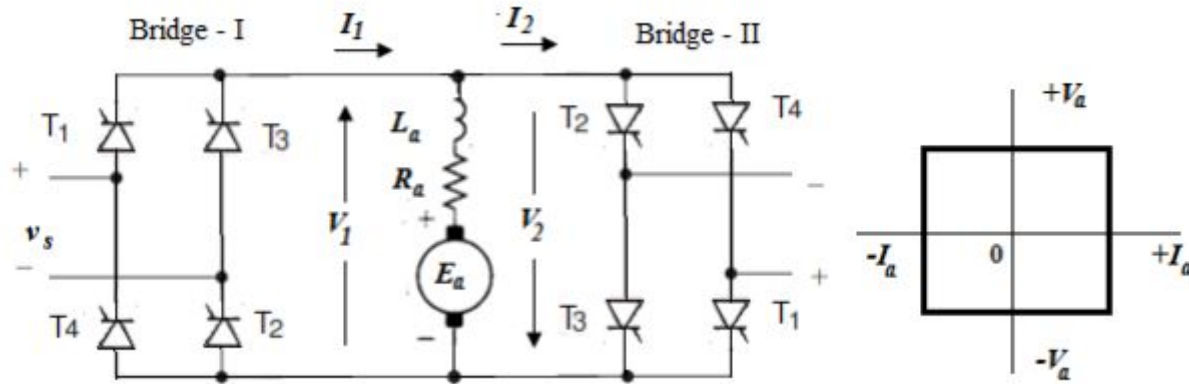
The choice of the converter implies the possibility of inverting the supply voltage, but not the armature current, therefore it will be possible to operate from motor in forward running and as reverse braking (**first and second quadrant**).

During the positive half wave T_1 and T_3 are directly polarized and conduct as soon as the gate pulse is sent to them. This conduction lasts until, in correspondence with the negative half-wave and the next gate pulse, T_2 and T_4 are commanded. The current is then switched and passed to this pair of devices, simultaneously bringing the previous pair of components to shutdown.





Single-phase dual rectifier



DC motors may require to be operated **in four quadrants** without a switching changeover. A simple dual converter drive which consists of **two single-phase full bridge converters** connected in inverse-parallel supplying a d.c motor is used. One bridge for one direction of motor current and the other bridge for the opposite direction of current. The controls are interlock to prevent their simultaneous operation to avoid short circuits on one another. **Bridge-I provides operation in the first and fourth quadrants while bridge-II provides operation in second and third quadrants.** This drive is employed for motors of rating up to 15 kW.



Three-phase full-wave half-controlled rectifier

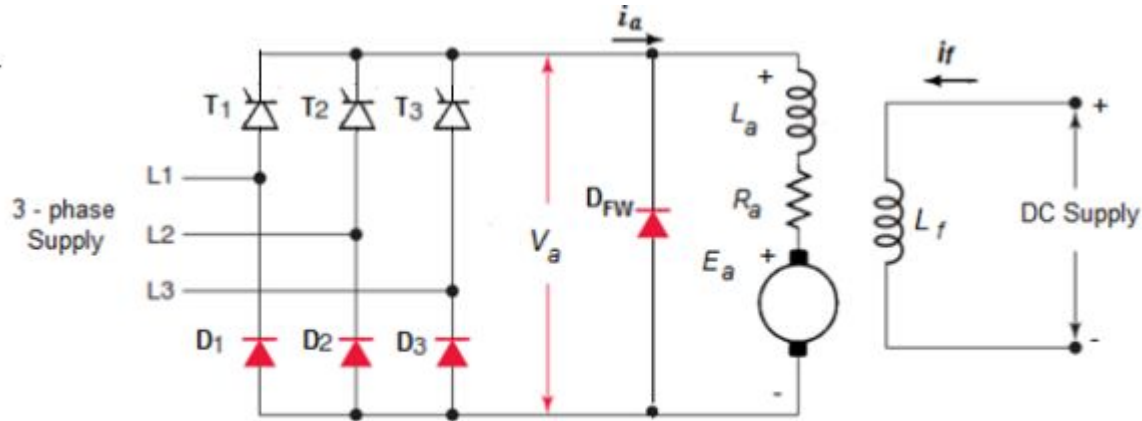


Fig.13.16 Three-phase semiconverter drive.

The three-phase semiconverter is a **one-quadrant drive**. Its circuit includes a freewheeling diode D_{FW} to maintain continuous load current. It uses three thyristors and three diodes; hence a cost advantage is obtained compared with the full-converter. This converter is used for motor ratings from 15 to 150 hp. The components activation sequence is:

$$T_1-D_2, T_2-D_3, T_3-D_1$$

A voltage is found on the load which will consist of "pieces" of the line-to-line voltages v_{AB} , v_{BC} , v_{CA} and $-v_{AB}$, $-v_{BC}$, $-v_{CA}$.



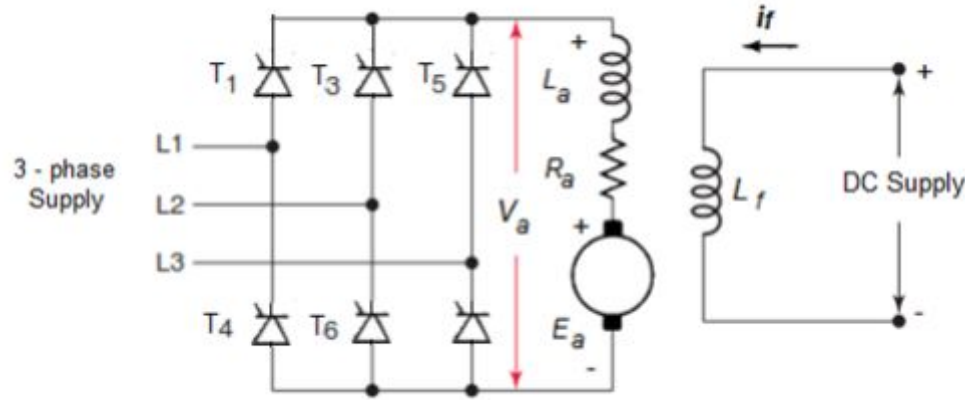
Three-phase full-wave half-controlled rectifier

- Since only three thyristors are used, the circuit is not expensive and a simple control circuitry is required;
- **Dynamic braking** can be performed by switching armature connection to an external resistance;
- Operation is **in the first quadrant** only;
- However, bi-directional rotation can be obtained by reversing field current or armature terminals when the motor has been stopped.

Since the devices are switched on with greater frequency, there will be a **lower ripple** of the voltage and current, with obvious advantages for the motor which will have a much **more regular torque** trend than in the case of the single-phase circuit. Another advantage is the **lower probability of extinction of the current** with consequent reduction of the occurrence of intermittent conduction.



Three-phase full-wave fully-controlled rectifier

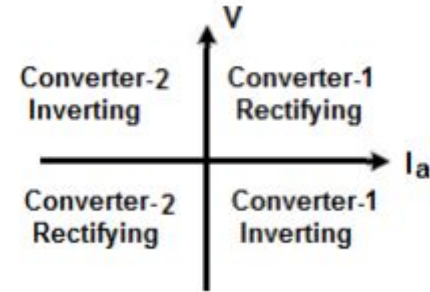
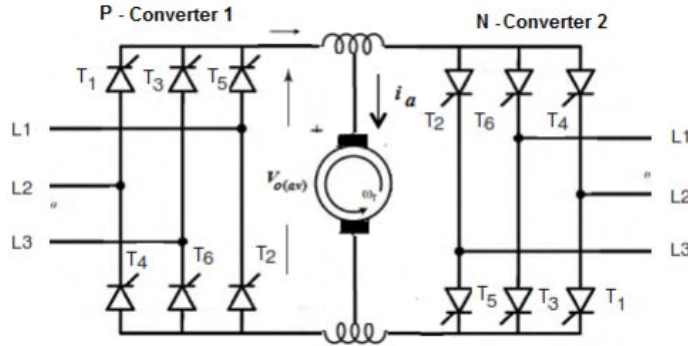


Advantages:

- **The load current is well balanced** across the bridge improving efficiency;
- **Ripple content is reduced**, both in amplitude and frequency, compared to the half-wave configuration.



Three-phase dual rectifier



Four-quadrant operation of a medium and large size d.c. motor drive (200-2000 hp) can be obtained. The average motor voltage is required to be equal for both converters, which implies that the firing angles of the two sets of the thyristors should sum to 180° .



DC Choppers



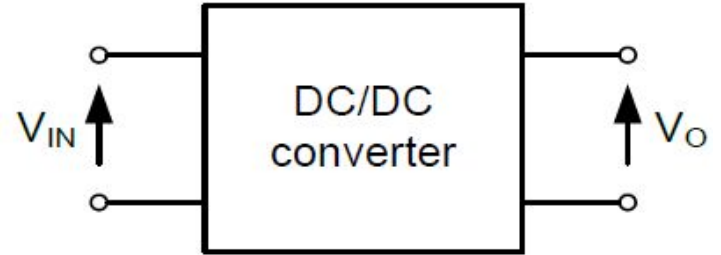
Chopper

A chopper is a static power electronic device that converts fixed DC input voltage to a variable DC output voltage. It could be considered as DC equivalent of an AC transformer since it behaves in an identical manner.

Chopper systems are characterized by:

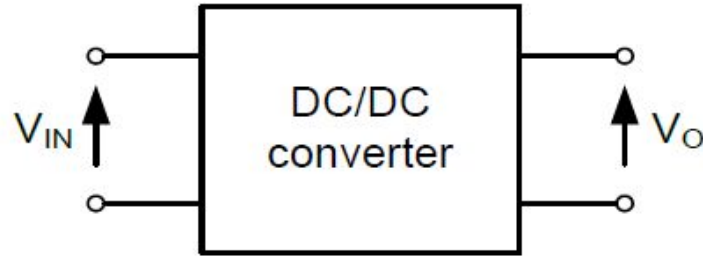
- high efficiency;
- fast response;
- regeneration operation capability.

The power semiconductor devices which are employed in these circuits can be force commutated thyristor, power BJT, MOSFET and IGBT. GTO and MCT based choppers are also used. All these devices actually operate like a switch.





Chopper - Classification



Choppers can be classified as follows:

- Class A (step-down or buck) □ I quadrant
- Class B (step-up or boost) □ IV quadrant
- Class C (inverting or buck-boost or half-bridge) □ I-IV quadrant
- Class D □ I-II quadrant
- Class E □ 4 quadrant



Chopper - principles



The chopper switch operates periodically with a chopper period equal to T , remains on for a time interval equal to T_{ON} and off for a time equal to T_{OFF} .

$$T = T_{ON} + T_{OFF}$$

It is possible to define the duty cycle as:

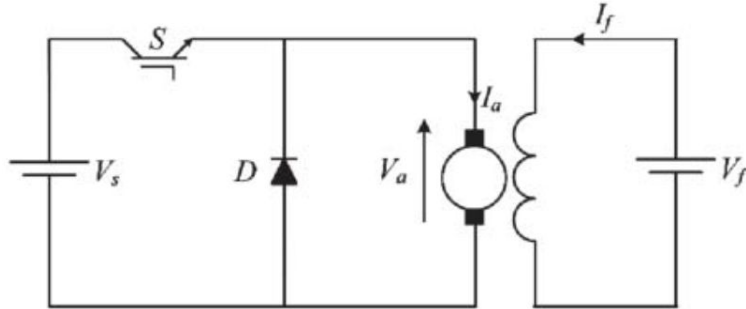
$$\delta = \frac{T_{ON}}{T} \quad 0 < \delta < 1$$

Pulse-width modulation (PWM) is adopted to control the armature voltage of DC motor drives for EV propulsion. Under the continuous conduction mode, the control is based on the variation of duty cycle δ of the power switch.



Chopper – Class A (step-down or buck)

The output voltage is always less than or at most equal to the supply voltage.



$$T_{ON} = \delta T$$

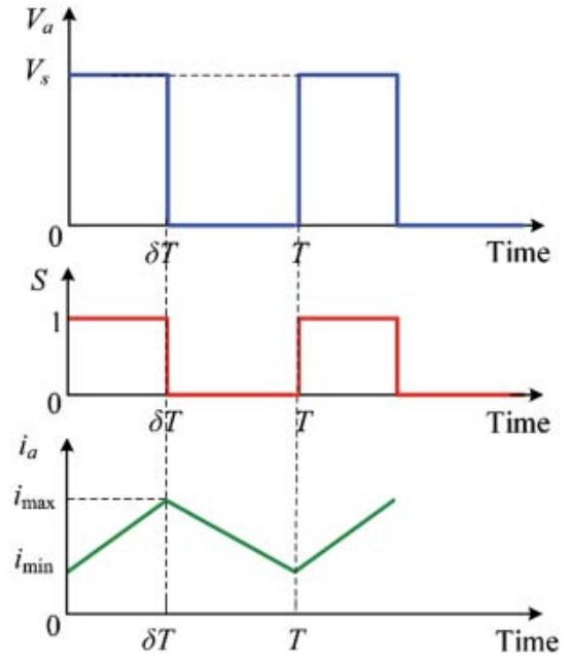
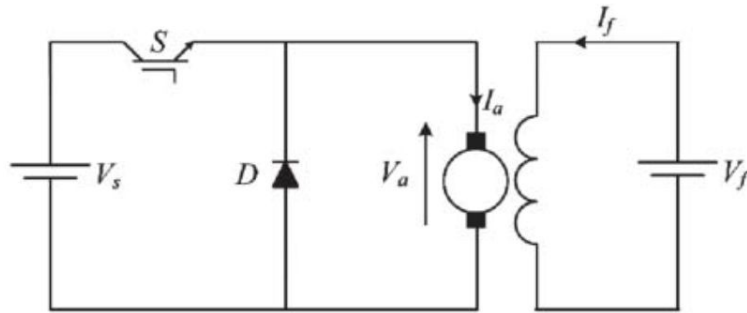
$$T_{OFF} = T - \delta T$$

In the interval equal to T_{ON} the switch is on and the voltage V is applied to the load (except for the drop on the chopper). As soon as the direct voltage V is applied to the ohmic-inductive load, the current has an exponential increasing trend. This trend is approximated with a straight line segment (linear).

$$\bar{V}_a = \frac{1}{T} \int_0^T V_a(t) dt = \delta V$$

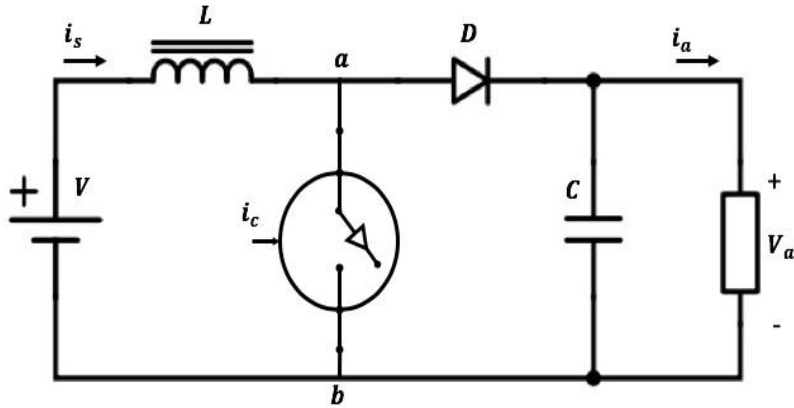


Chopper – Class A (step-down or buck)





Chopper – Class B (step-up or boost)



$$V_a = \frac{V}{1 - \delta}$$

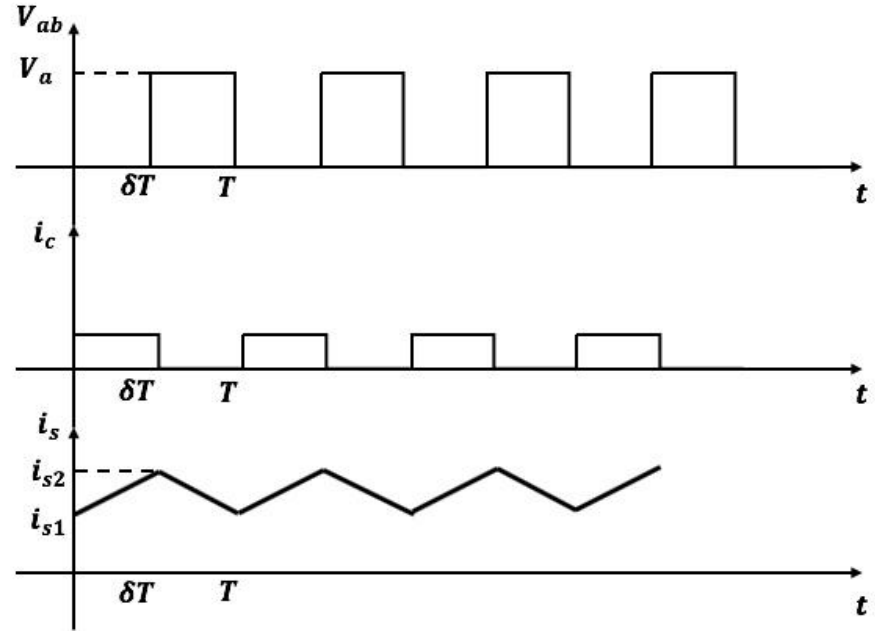
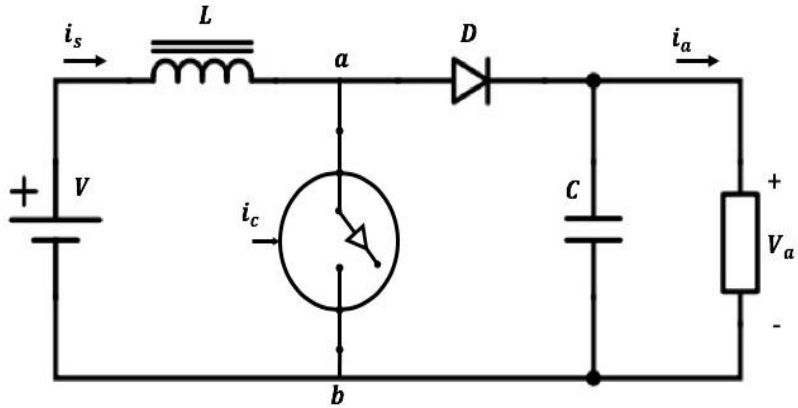
The output voltage is greater than the input voltage

When the switch is turned on, **the power supply supplies energy to the inductance** which accumulates it and there is a growth of the current. The previously loaded capacity cannot be discharged on the power supply due to the diode and is discharged on the motor where **the load current tends to decrease** due to the dissipative energy balance.

Subsequently, the switch is turned off and **the energy previously stored on the inductance is transferred onto the capacity** and the power source (but mainly on the capacity) while the current on the power supply side therefore tends to decrease.

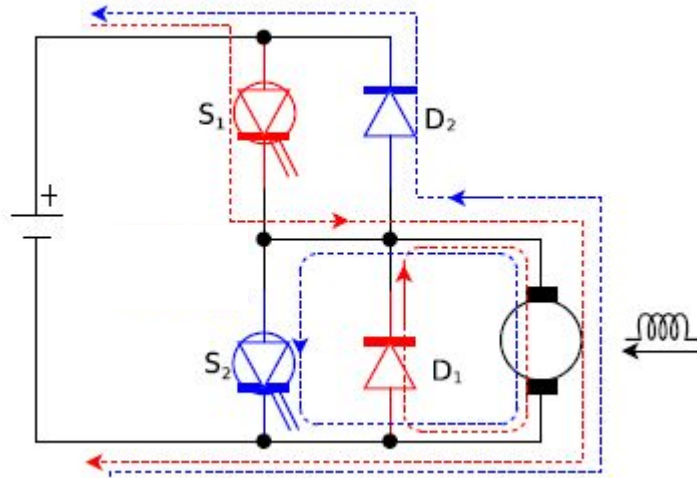


Chopper – Class B (step-up or boost)



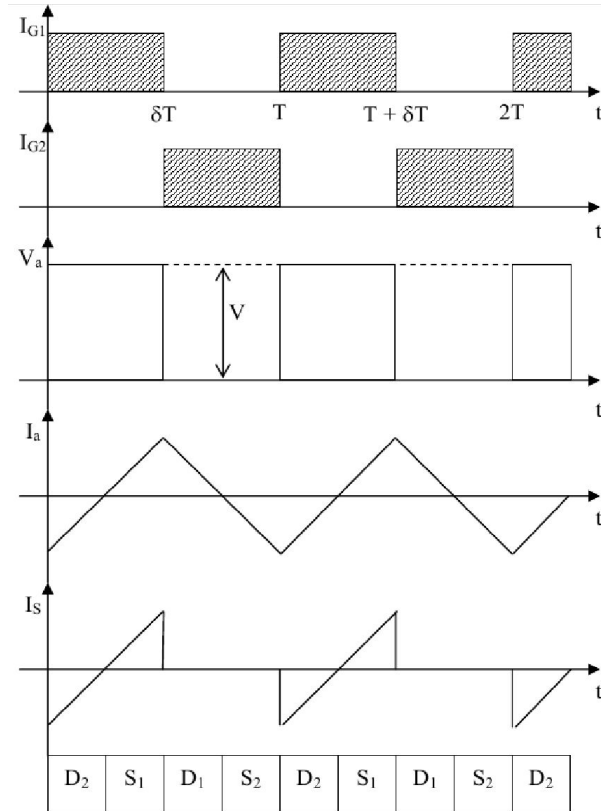


Chopper – Class C (inverting or buck-boost or half-bridge)



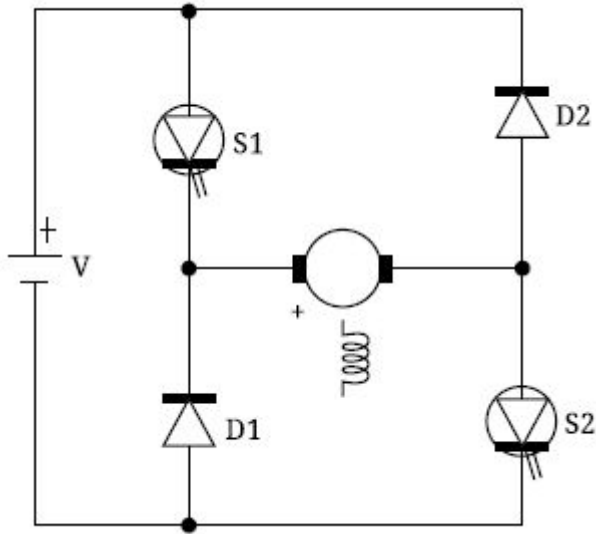
S_1 D_1 realizes the function of the buck chopper, while the pair S_2 D_2 realizes the function of boost chopper.

$$V_a = \frac{V\delta}{1 - \delta}$$





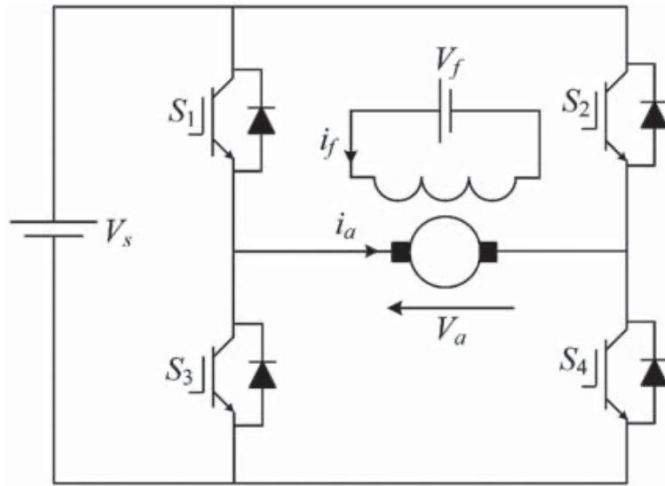
Chopper – Class D



- S_1 and S_2 in conduction, the voltage V_a is equal to V ; (motor)
- S_1 and D_2 in conduction (S_2 is off) and obviously $V_a = 0$; (recirculation)
- S_2 and D_1 in conduction (S_1 is disabled) and also in this case $V_a = 0$; (recirculation)
- D_1 and D_2 in conduction (with S_1 and S_2 both off). (brake)



Chopper – Class E (Full-bridge)



Motoring and regenerative braking in both EV forward and backward operations are controlled electronically by means of PWM, that is, by varying the duty cycle.

The four-quadrant DC chopper can offer the positive armature voltage with bidirectional armature current and also the negative armature voltage with bidirectional armature current, hence achieving **forward motoring** and **forward regenerative braking** as well as **backward motoring** and **backward regenerative braking**.

CONTACTS

Fabrizio Donatantonio

fabrizio.donatantonio@kineton.it

Giuseppe Galioto

giuseppe.galioto@kineton.it

Via E. Gianturco, 23
(presso complesso Mecfond)
80146 Napoli

www.kineton.it

